

Christine Leanna Bonoan

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Education

CALIFORNIA STATE UNIVERSITY OF BAKERSFIELD

Post-baccalaureate | B.S., Computer Science

Bakersfield, CA

Graduated: May 2026

UNIVERSITY OF CALIFORNIA IRVINE

B.A., Criminology, Law and Society

Irvine, CA

Graduated: June 2022

Projects and Activities

SENIOR DESIGN CAPSTONE: Shadow-Based Image Tampering Detection System

<https://github.com/clbonoan/senior-project>

- Developed a **full-stack web application** (Flask, Python, OpenCV, NumPy, JavaScript, HTML/CSS) to identify digital manipulation through shadow analysis
- Programmed **3 Python**-based shadow-consistency detection algorithms (**OpenCV** and **NumPy**) to extract texture, geometric, and illumination-based features
- Built a comparison framework to evaluate **rule**-based detection against **ML**-based scoring approaches

COMPUTER NETWORKS: Smart Lighting System Efficiency Optimization

<https://github.com/clbonoan/energy-efficiency>

- Directed a 5-person team project to design and implement an automated smart lighting system using a secure **MQTT**-based publish/subscribe communication architecture
- Wrote **Python** scripts to control lighting based on real-time motion and ambient light sensor data, and to log power consumption and system events for performance analysis
- Yielded an **81% reduction in energy consumption** by adding contextual triggers, documented in an **8-page** paper

DATABASE SYSTEMS: 'Otaku Clash' Website

<https://artemis.cs.csub.edu/~otakuclash/index.php>

- Collaborated with a 4-person team to design a role-based web application with user/admin authentication and over **6** interactive features
- Developed a **MySQL relational database** that implemented complex joins and views to dynamically generate leaderboards, suggested characters, and admin tools for user and content management
- Enhanced data reliability and improved query performance through optimization and validation

SOFTWARE ENGINEERING: C++ Game Development

<https://github.com/clbonoan/space-busters>

- Developed a modular **C++** game application using object-oriented design principles, real-time update loops, and event-driven state management
- Implemented collision detection, input handling, and gameplay state transitions to support responsive player actions
- Diagnosed and resolved a critical bug affecting loop execution across state transitions, improving program control flow

DISTRIBUTED AND PARALLEL COMPUTATION: Murder Mystery Gossip Simulation

<https://github.com/clbonoan/murder-mystery>

- Collaborated on a **Rust**-based distributed simulation where nodes exchanged gossip messages and heartbeat updates across decentralized processes
- Contributed to **protobuf**-based messaging, node communication logic, and gossip propagation across nodes
- Helped integrate WebSocket communication, cloud deployment workflows, and testing using **AWS**, **Docker**, and **SST**

HACKATHON: CSU Channel Islands Plot-A-Thon - Data Visualization

Best Overall - Best in Data Analysis - Best in Data Communication

- Participated with a 4-person team to clean and analyze a large-scale, nationally representative survey dataset (Gamer Study 2024) using **Pandas**
- Created **Matplotlib** visualizations and an infographic to communicate trends and actionable insights on music discovery through video games

Skills

Languages: Python, C++, HTML, CSS, Javascript
Frameworks/Libraries: OpenCV, NumPy, Pandas, Flask

Databases: SQL (MySQL)
Systems/Networking: Linux, MQTT, Zigbee

Tools: Git